How Plants Make Food: Photosynthesis

Plants Are Food Makers

Animals have to find their food, but green plants can make their own. They do it inside their leaves, using a process called photosynthesis. The word photosynthesis means "putting together with light." Almost every living thing on Earth depends on this amazing trick.

What Plants Need

To make food, a plant needs three things: water, carbon dioxide and sunlight. The roots soak up water from the soil, and tiny tubes carry it up the stem to the leaves. Carbon dioxide is a gas in the air. It enters the leaf through tiny holes called stomata. Sunlight is the energy source. It powers the whole process, like a battery powers a toy.

How Leaves Turn Light into Sugar

Leaves are green because they are full of chlorophyll. Chlorophyll is a green pigment that traps the energy in sunlight. The leaf uses this light energy to join water and carbon dioxide together. This makes a kind of sugar called glucose. Glucose is the plant's food. The plant uses it to grow new leaves, stems, roots, flowers and fruit, and it can store extra sugar as starch for later.

Oxygen: The Gift Plants Give Us

When a plant makes sugar, it also makes oxygen. The plant does not need all of this oxygen, so it lets it out into the air through the stomata. People and animals need oxygen to breathe. Every time you take a breath, you are breathing oxygen that plants helped to make.

Why Photosynthesis Matters

Photosynthesis is the start of almost every food chain. A rabbit eats grass, and a fox eats the rabbit, so the energy in the fox's food first came from sunlight caught by a plant. Plants also take carbon dioxide out of the air. Without photosynthesis, there would be no food for animals and not enough oxygen to breathe.

Quick Recap

Water plus carbon dioxide plus light energy make sugar and oxygen. Remember it in three steps. First, the plant collects water and carbon dioxide. Next, chlorophyll catches the sunlight. Last, the leaf makes sugar and releases oxygen into the air.